

Laboratório 07: SQLite II

AUTHOR

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Questões

1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
library(RSQLite)

conn <- dbConnect(
  SQLite(),
  "uwmadison.sqlite3"
)
dbListTables(conn)
```

```
[1] "course_offerings"  "courses"           "grade_distributions"
[4] "instructors"       "rooms"              "schedules"
[7] "sections"         "subject_memberships" "subjects"
[10] "teachings"
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT'). Quem são os professores?

```
SELECT DISTINCT
  instructors.id,
  instructors.name
FROM subjects
INNER JOIN subject_memberships
  ON subjects.code = subject_memberships.subject_code
INNER JOIN course_offerings
  ON subject_memberships.course_offering_uuid = course_offerings.uuid
INNER JOIN sections
  ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN teachings
  ON sections.uuid = teachings.section_uuid
INNER JOIN instructors
  ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT'
ORDER BY instructors.name;
```

Displaying records 1 - 10

id	name
4787911	ABIGAIL BENZINE
3111842	ADIN-CRISTIAN ANDREI
3775809	AKICHIKA OZEKI

id	name
4958796	ALAN HUANG
5451706	ALBERTO DEL PIA
3661782	ALBRECHT KLEMM
5656034	ALEXANDER COVINGTON
4680082	ALEXANDER FISH
3573775	ALEXANDER KISELEV
5162339	ALYSSA DIGILIO

Quantos professores são?

```
SELECT COUNT(DISTINCT instructors.id) AS quantidade_professores
FROM subjects
INNER JOIN subject_memberships
    ON subjects.code = subject_memberships.subject_code
INNER JOIN course_offerings
    ON subject_memberships.course_offering_uuid = course_offerings.uuid
INNER JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN teachings
    ON sections.uuid = teachings.section_uuid
INNER JOIN instructors
    ON teachings.instructor_id = instructors.id
WHERE subjects.abbreviation = 'STAT';
```

1 records

quantidade_professores
379

3.O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

###Para encontrar o professor associado ao **menor GPA médio**

```
WITH gpas AS (
    SELECT
        course_offering_uuid,

        (
            a_count * 4.0 +
            ab_count * 3.5 +
            b_count * 3.0 +
            bc_count * 2.5 +
            c_count * 2.0 +
            d_count * 1.0 +
            f_count * 0.0
```

```
    )
    /
    NULLIF(
        a_count +
        ab_count +
        b_count +
        bc_count +
        c_count +
        d_count +
        f_count,
        0
    ) AS gpa

FROM grade_distributions
)

SELECT
    instructors.id,
    instructors.name,
    AVG(gpas.gpa) AS gpa_medio

FROM gpas
INNER JOIN course_offerings
    ON gpas.course_offering_uuid = course_offerings.uuid
INNER JOIN sections
    ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN teachings
    ON sections.uuid = teachings.section_uuid
INNER JOIN instructors
    ON teachings.instructor_id = instructors.id
INNER JOIN subject_memberships
    ON course_offerings.uuid = subject_memberships.course_offering_uuid
INNER JOIN subjects
    ON subject_memberships.subject_code = subjects.code

WHERE subjects.abbreviation = 'STAT'

GROUP BY
    instructors.id,
    instructors.name

ORDER BY
    gpa_medio ASC

LIMIT 1;
```

1 records

id	name	gpa_medio
138086	CHRISTINE SORKNESS	NA

###Para encontrar o professor associado ao **maior GPA médio**

```

WITH gpas AS (
  SELECT
    course_offering_uuid,

    (
      a_count * 4.0 +
      ab_count * 3.5 +
      b_count * 3.0 +
      bc_count * 2.5 +
      c_count * 2.0 +
      d_count * 1.0 +
      f_count * 0.0
    )
    /
    NULLIF(
      a_count +
      ab_count +
      b_count +
      bc_count +
      c_count +
      d_count +
      f_count,
      0
    ) AS gpa

  FROM grade_distributions
)

SELECT
  instructors.id,
  instructors.name,
  AVG(gpas.gpa) AS gpa_medio

FROM gpas
INNER JOIN course_offerings
  ON gpas.course_offering_uuid = course_offerings.uuid
INNER JOIN sections
  ON course_offerings.uuid = sections.course_offering_uuid
INNER JOIN teachings
  ON sections.uuid = teachings.section_uuid
INNER JOIN instructors
  ON teachings.instructor_id = instructors.id
INNER JOIN subject_memberships
  ON course_offerings.uuid = subject_memberships.course_offering_uuid
INNER JOIN subjects
  ON subject_memberships.subject_code = subjects.code

WHERE subjects.abbreviation = 'STAT'

GROUP BY
  instructors.id,
  instructors.name

ORDER BY

```

gpa_medio DESC

LIMIT 1;

1 records

id	name	gpa_medio
3884290	SRIKANTHMADHAVAN ARAVAMUTHAN	4

###Para as disciplinas

```
WITH gpas AS (  
  SELECT  
    course_offering_uuid,  
  
    (  
      a_count * 4.0 +  
      ab_count * 3.5 +  
      b_count * 3.0 +  
      bc_count * 2.5 +  
      c_count * 2.0 +  
      d_count * 1.0 +  
      f_count * 0.0  
    )  
    /  
    NULLIF(  
      a_count +  
      ab_count +  
      b_count +  
      bc_count +  
      c_count +  
      d_count +  
      f_count,  
      0  
    ) AS gpa  
  
  FROM grade_distributions  
)  
  
SELECT  
  courses.number,  
  courses.name,  
  AVG(gpas.gpa) AS gpa_medio  
  
FROM gpas  
  
INNER JOIN course_offerings  
  ON gpas.course_offering_uuid =  
     course_offerings.uuid  
  
INNER JOIN courses  
  ON course_offerings.course_uuid =
```

```
courses.uuid

INNER JOIN subject_memberships
  ON course_offerings.uuid =
     subject_memberships.course_offering_uuid

INNER JOIN subjects
  ON subject_memberships.subject_code =
     subjects.code

WHERE subjects.abbreviation = 'STAT'

GROUP BY
  courses.uuid,
  courses.number,
  courses.name

ORDER BY
  gpa_medio;
```

Displaying records 1 - 10

number	name	gpa_medio
698	Directed Study	NA
546	Practicum in Clinical Trial Data Analysis and Interpretation	NA
990	Research	NA
768	Statistical Methods for Medical Image Analysis	NA
699	Directed Study	NA
681	Senior Honors Thesis	NA
682	Senior Honors Thesis	NA
431	Introduction to the Theory of Probability	2.891568
525	Linear Programming Methods	2.957360
309	Introduction to Probability and Mathematical Statistics I	3.031951

A disciplina com o menor GPA médio

```
WITH gpas AS (
  SELECT
    course_offering_uuid,

    (
      a_count * 4.0 +
      ab_count * 3.5 +
      b_count * 3.0 +
      bc_count * 2.5 +
      c_count * 2.0 +
      d_count * 1.0 +
      f_count * 0.0
```

```
)
/
NULLIF(
    a_count +
    ab_count +
    b_count +
    bc_count +
    c_count +
    d_count +
    f_count,
    0
) AS gpa

FROM grade_distributions
)

SELECT
    courses.number,
    courses.name,
    AVG(gpas.gpa) AS gpa_medio

FROM gpas
INNER JOIN course_offerings
    ON gpas.course_offering_uuid = course_offerings.uuid
INNER JOIN courses
    ON course_offerings.course_uuid = courses.uuid
INNER JOIN subject_memberships
    ON course_offerings.uuid = subject_memberships.course_offering_uuid
INNER JOIN subjects
    ON subject_memberships.subject_code = subjects.code

WHERE subjects.abbreviation = 'STAT'

GROUP BY
    courses.uuid,
    courses.number,
    courses.name

ORDER BY
    gpa_medio ASC

LIMIT 1;
```

1 records

number	name	gpa_medio
698	Directed Study	NA

disciplina com o maior GPA médio

```
WITH gpas AS (
    SELECT
```

```
        course_offering_uuid,

        (
            a_count * 4.0 +
            ab_count * 3.5 +
            b_count * 3.0 +
            bc_count * 2.5 +
            c_count * 2.0 +
            d_count * 1.0 +
            f_count * 0.0
        )
        /
        NULLIF(
            a_count +
            ab_count +
            b_count +
            bc_count +
            c_count +
            d_count +
            f_count,
            0
        ) AS gpa

    FROM grade_distributions
)

SELECT
    courses.number,
    courses.name,
    AVG(gpas.gpa) AS gpa_medio

FROM gpas
INNER JOIN course_offerings
    ON gpas.course_offering_uuid = course_offerings.uuid
INNER JOIN courses
    ON course_offerings.course_uuid = courses.uuid
INNER JOIN subject_memberships
    ON course_offerings.uuid = subject_memberships.course_offering_uuid
INNER JOIN subjects
    ON subject_memberships.subject_code = subjects.code

WHERE subjects.abbreviation = 'STAT'

GROUP BY
    courses.uuid,
    courses.number,
    courses.name

ORDER BY
    gpa_medio DESC

LIMIT 1;
```


1 records

number	name	gpa_medio
841	Nonparametric Statistics and Machine Learning Methods	4

4.Desconecte do banco de dados.

```
dbDisconnect(conn)
```